Decision letter (Initial Submission)

TPAMI-2025-10-3142 - Major Revision Required

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CC: kyoungmu@gmail.com

RE: TPAMI-2025-10-3142, "Enhancing Multi-Robot Goal-Oriented Navigation Through Multimodal Chain-of-Thought Score Collaboration"
Manuscript Type: Regular

Dear Dr. Zhixuan Shen,

We have completed the review process of the above referenced paper for the IEEE Transactions on Pattern Analysis and Machine Intelligence and recommend that your paper undergo a Major Revision.

Enclosed are your reviews. If you should choose to revise your paper, please prepare a separate document describing how each of the reviewers' comments are responded to in your revision and send it to us by 19-Feb-2026.

When you are ready to submit your revision, visit the following link:
https://ieee.atyponrex.com/submission/submissionBoard/REX-PROD-2-BD5EB7FC-AC78-48F2-A655-E30D88664C91-8758D917-17F9-41CC-9A2C-110C10F94E79-85407/current?idtype=external

Once the revised manuscript is prepared, you can upload it and submit it through your Author Center.

When submitting your revised manuscript, you will be able to respond to the comments made by the reviewer(s) in the space provided. You can use this space to document any changes you make to the original manuscript. In order to expedite the processing of the revised manuscript, please be as specific as possible in your response to the reviewer(s)' questions and comments. You may also upload your responses as separate files for review along with your revision. If you choose to do this, please choose "Summary of Changes" as the file designation.

When the submission process is complete, you will receive an automated confirmation email immediately. If you did not receive that email, your submission is not yet complete.

I will contact you should we have any concerns or questions regarding your revision. Otherwise, your revision will be forwarded to the assigned Associate Editor with a request to begin the second round of reviews.

Please be mindful when making your revisions that you still need to maintain the size limitations for papers submitted to TPAMI. Our manuscript types and submission length guidelines (including the main text, the abstract, index terms, illustrations and references) are found at,

https://www.computer.org/csdl/journal/tp/write-for-us/15083?title=Author%20Information&periodical=IEEE%20Transactions%20on%20Pattern%20Analysis%20and%20Machine%20Intelligence

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Important: Your main manuscript file cannot include colored or highlighted text. Please upload a clean, publication-ready version of your manuscript under the "Formatted (Double Column) Main File - PDF Document Only" file designation. If you would like to include an annotated version of your main manuscript file, please upload it under the "Summary of Changes" file designation.

Your revised paper must include the following:

- Abstract
- Index terms
- Author affiliation information
- Main text
- References
- Figure captions
- Table titles
- Brief biography of each author
(biographies are not required for concise papers or comments papers)

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irrelevant, please note your concerns in your response to reviewer feedback.

Please do not hesitate in contacting us should you have any questions about our process or are experiencing technical difficulties. You can reach me at jarnold@computer.org.

Thank you for your contribution to TPAMI, and we look forward to receiving your revised manuscript.

Thank you,

Mrs. Joyce Arnold, Administrator on behalf of Dr. Kyoung Mu Lee (EIC), EIC
IEEE Transactions on Pattern Analysis and Machine Intelligence
jarnold@computer.org

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Editor Comments

Associate Editor
Comments to the Author:
The paper proposes an approach for multi-robot goal-oriented navigation in unseen environments. The paper was reviewed by three experts in the field. While they appreciate the well-designed approach, they raised concerns regards the use of LMMs, presentation, e.t.c. The authors are encouraged to fully address these concerns.

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Reviewers' Comments

Please note that some reviewers may have included additional comments in a separate file. If a review contains the note "see the attached file" under Section III A - Public Comments, you will need to log on to the submission site to view the file. After logging in, select the Author Center, click on the "Manuscripts with Decisions" queue and then clicking on the "view decision letter" link for this manuscript. You must scroll down to the very bottom of the letter to see the file(s), if any. This will open the file that the reviewer(s) or the Associate Editor included for you along with their review.

Reviewer: 1

Comments:
This paper investigates the multi-robot goal-oriented navigation (MRON) task and proposes a discussion-driven chain-of-thought navigation framework (DisCoNav). The authors introduce several engineering-oriented modules, including an initial target deliberation module, an exploration and backtracking judgment module, and an adversarial consensus optimization module. Overall, however, I believe the novelty is limited, and the current technical depth does not meet TPAMI standards. My detailed comments are as follows:

1. Comments on the Introduction
(1) The statement “However, in complex and diverse indoor scenes, CoT-based navigation methods often prove unreliable due to meaningless invalid communication messages, as illustrated in Figure 1(a)” does not clearly correspond to Figure 1. It is unclear which existing methods exhibit this issue and how frequently it occurs.
(2) In Page X, Paragraph 5, the MRON task is introduced without appropriate citations.
(3) As this work is an extended version of an AAAI 2025 paper, the authors should explicitly clarify the key innovations of the conference version, what additional contributions are made in this journal submission, and how the extended methods improve the results.
(4) The contributions claim that the proposed modules are low-cost and address communication inefficiency. However, these advantages are not supported by the subsequent descriptions or experimental evidence. In fact, the method significantly increases runtime (e.g., over 10 minutes per episode with a single robot and up to 20 minutes with three robots). The stated contributions appear inconsistent with the actual results.

2. Comments on the Method
(1) Prior work [1] has shown that LVLMs often struggle with accurate spatial understanding. In Sec. IV-B, the authors rely on LVLMs to describe spatial and positional information. Has the correctness of these outputs been validated? Do they introduce additional noise?
(2) For N robots, how are the thinker and refuter roles assigned? Figure 6 states that a refuter is chosen randomly; the impact of such randomness on performance should be analyzed and discussed.
(3) In Sec. IV-D, some long-range targets are not reached. The authors should explain why these failures occur.
[1] "Ecbench: Can multi-modal foundation models understand the egocentric world? a holistic embodied cognition benchmark." Proceedings of the Computer Vision and Pattern Recognition Conference. 2025.

3. Additional Concerns Regarding Experiments

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Additional Questions:
1. Which category describes this manuscript?: Research/Technology

2. How relevant is this manuscript to the readers of this periodical? If you answer Not very relevant or Irrelevant please explain your rating under Public Comments below.: Very Relevant

1. Please evaluate the significance of the manuscript's research contribution.: Fair - Even with the recommended changes, the contribution of this paper is unlikely be significant enough for publication in TPAMI.

2. Please explain how this manuscript advances this field of research and/or contributes something new to the literature.: Regarding how this manuscript advances the field, I have reservations. Although the paper studies the MRON task and introduces a discussion-driven chain-of-thought framework (DisCoNav) with several engineered modules, these components do not appear sufficiently novel or distinctive. Moreover, they introduce substantial latency, making the system slow and limiting its practicality and scalability. Thus, the manuscript's contribution to advancing the literature remains unclear.

3. Is the manuscript technically sound? In the Public Comments section, please provide detailed explanations to support your assessment: Partially

1. Are the title, abstract, and keywords appropriate? If not, please comment in the Public Comments section.: Yes

4. How thorough is the experimental validation (where appropriate)? Please discuss any shortcomings in the Public Comments section.: Lacking in some respects; some cases of interest not tested

2. Does the manuscript contain sufficient and appropriate references? Please comment and include additional suggested references in the Public Comments section.: Important references are missing; more references are needed

If you are suggesting additional references they must be entered in the text box provided. All suggestions must include full bibliographic information plus a DOI.

If you are not suggesting any references, please type NA.: Dang, Ronghao, et al. "Ecbench: Can multi-modal foundation models understand the egocentric world? a holistic embodied cognition benchmark." Proceedings of the Computer Vision and Pattern Recognition Conference. 2025. DOI: 10.1109/CVPR52734.2025.02290

3. Does the introduction state the objectives of the manuscript in terms that encourage the reader to read on? If not, please explain your answer in the Public Comments section.: Could be improved

4. How would you rate the organization of the manuscript? Is it focused? Please elaborate with suggestions for reorganization in the Public Comments section.: Satisfactory

6. How is the length of the manuscript? If changes are suggested, please make explicit recommendations in the Public Comments section.: About right

5. Please rate the readability of the manuscript. Explain your rating under Public Comments below.: Readable - but requires some effort to understand

7. Should the supplemental material be included? (Click on the Supplementary Files icon to view files): Does not apply, no supplementary files included

8. If yes to 7, should it be accepted:

Please rate the manuscript overall. Explain your choice.: Fair

Reviewer: 2

Comments:
1. The introduction primarily contrasts centralized and decentralized planning paradigms for multi-robot navigation. While this provides useful background, could the authors clarify whether this dichotomy fully reflects the current design space of multi-agent coordination? In recent years, hybrid strategies—such as centralized training with decentralized execution [1], or structured and learned communication [2]—have been explored to balance scalability, communication efficiency, and coordination. Given that DisCoNav employs a global semantic map alongside decentralized execution, it may be helpful to explicitly discuss its relationship to these hybrid paradigms.

2. The method extensively leverages LVLM-generated multimodal chain-of-thought prompts. However, in the current

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better position the methodological contribution.

4. In the later stages of the adversarial consensus optimization, the final long-term target allocation is solved via a standard VCG-based combinatorial auction with a centralized winner determination process. Since the auction operates solely on upstream valuation scores, could the authors clarify how this mechanism relates to the claimed adversarial discussion, and to what extent the final coordination outcome depends on multi-agent reasoning versus classical auction-based coordination?

5. Most baselines in the experiments focus on heuristic or non-collaborative strategies. Given the emphasis on adversarial discussion and consensus optimization, could the authors include or discuss stronger baselines that use the same coordination backbone (e.g., voting or auction) but without adversarial discussion or chain-of-thought reasoning, in order to better isolate the contribution of the proposed components?

6. The GT-Seg experiments highlight the strong dependence of navigation performance on semantic segmentation quality. It would be helpful to discuss how this dependence interacts with the proposed reasoning and discussion modules, and whether their relative contribution diminishes as perception quality improves.

7. The ablation results show that removing the discussion component can sometimes improve performance, and that increasing the number of discussion rounds consistently degrades both SPL and SR while incurring substantial computational overhead. Could the authors clarify under what conditions adversarial discussion is truly beneficial, and whether it should be considered a core contributor to the overall performance gains?

Reference:
[1] Rashid, Tabish, et al. "Monotonic value function factorisation for deep multi-agent reinforcement learning." Journal of Machine Learning Research 21.178 (2020): 1-51.
[2] Sheng, Junjie, et al. "Learning structured communication for multi-agent reinforcement learning." Autonomous Agents and Multi-Agent Systems 36.2 (2022): 50.

Additional Questions:
1. Which category describes this manuscript?: Research/Technology

2. How relevant is this manuscript to the readers of this periodical? If you answer Not very relevant or Irrelevant please explain your rating under Public Comments below.: Relevant

1. Please evaluate the significance of the manuscript's research contribution.: Good

2. Please explain how this manuscript advances this field of research and/or contributes something new to the literature.: This manuscript contributes to the literature by providing a system-level exploration of integrating large vision–language models into multi-robot goal-oriented navigation. The proposed framework combines LVLM-based decision scoring with shared semantic mapping and auction-based coordination, and evaluates the resulting system on standard multi-robot navigation benchmarks.

Although many individual components build on existing techniques, the work offers empirical evidence on how LVLM-based scoring and structured coordination mechanisms interact in multi-robot settings, which may be useful for informing future designs of hybrid reasoning and coordination systems.

3. Is the manuscript technically sound? In the Public Comments section, please provide detailed explanations to support your assessment: Appears to be - but didn't check completely

1. Are the title, abstract, and keywords appropriate? If not, please comment in the Public Comments section.: Yes

4. How thorough is the experimental validation (where appropriate)? Please discuss any shortcomings in the Public Comments section.: Lacking in some respects; some cases of interest not tested

2. Does the manuscript contain sufficient and appropriate references? Please comment and include additional suggested references in the Public Comments section.: Important references are missing; more references are needed

If you are suggesting additional references they must be entered in the text box provided. All suggestions must include full bibliographic information plus a DOI.

If you are not suggesting any references, please type NA.: NA

3. Does the introduction state the objectives of the manuscript in terms that encourage the reader to read on? If not, please explain your answer in the Public Comments section.: Could be improved

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5. Please rate the readability of the manuscript. Explain your rating under Public Comments below.: Easy to read

7. Should the supplemental material be included? (Click on the Supplementary Files icon to view files): Does not apply, no supplementary files included

8. If yes to 7, should it be accepted:

Please rate the manuscript overall. Explain your choice.: Fair

Reviewer: 3

Comments:
Summary

This paper studies multi-robot goal-oriented navigation in unseen environments, with the goal of reducing conflicts and redundant exploration. The authors propose a system that combines local multimodal reasoning, a shared semantic map, a discussion phase among robots, and a final consensus step to decide where each robot should go. The paper presents a fairly complex but coherent pipeline. In this work, each robot first makes its own decision, then robots exchange information to adjust those decisions, and finally they agree on who explores which area to avoid conflicts. The system is carefully engineered and achieves strong zero-shot performance on the HM3D v0.2 and MP3D datasets.

Strengths

1.The idea is intuitive and make sense. The high-level pipeline makes sense: individual reasoning first, coordination next, and consensus at the end. Using a shared semantic map as the main communication medium feels like a reasonable design choice.

2.The method is carefully implemented and detailed. The paper provides many technical details, which makes the approach reproducible.

3.The proposed method consistently outperforms several baselines on standard benchmarks. The improvements are meaningful, and the ablation studies generally support the usefulness of the key components such as discussion and consensus.

Weaknesses

1.The method section could be easier to read. While the method is described in great technical detail, the section is quite long and often mixes high-level ideas with low-level implementation details. As a result, it can be hard to quickly understand the main logic of each module. The paper would likely be easier to read if each subsection started with a short high-level summary that explains the main idea and data flow before going into details. Some very specific implementation choices (for example, how robot history nodes are indexed using lowercase letters and extended numbering when the number exceeds 26) would fit better in an implementation details section or in the supplementary material, rather than in the main method.

2.The system design is currently explained in great detail inside the method section. While this information is useful, it interrupts the main algorithmic flow. It may be better for the authors to show concrete prompt templates, formatting rules, and response examples as part of the supplementary material, while keeping the high-level reasoning strategy in the main paper.

3.The paper mentions that states that centralized strategies suffer from exponentially growing computational cost as the environment and number of robots increase. It would strengthen the paper to support this claim with quantitative comparisons under the same setting in experiment.

4.As shown in Table IV, the average episode completion time increases significantly as the number of robots grows. It is unclear whether this level of runtime cost would be acceptable in practice, especially for scenarios with more than three robots. Some discussion or analysis of scalability would be helpful.

5.Providing a few concrete examples in the supplementary material showing how the refuter–thinker interaction helps correct wrong decisions would make the discussion mechanism more convincing.

Additional Questions:
1. Which category describes this manuscript?: Research/Technology

2. How relevant is this manuscript to the readers of this periodical? If you answer Not very relevant or Irrelevant please

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performance in zero-shot, multi-robot exploration.

3. Is the manuscript technically sound? In the Public Comments section, please provide detailed explanations to support your assessment: Appears to be - but didn't check completely

1. Are the title, abstract, and keywords appropriate? If not, please comment in the Public Comments section.: Yes

4. How thorough is the experimental validation (where appropriate)? Please discuss any shortcomings in the Public Comments section.: Compelling experiments; clearly state of the art

2. Does the manuscript contain sufficient and appropriate references? Please comment and include additional suggested references in the Public Comments section.: References are sufficient and appropriate

If you are suggesting additional references they must be entered in the text box provided. All suggestions must include full bibliographic information plus a DOI.

If you are not suggesting any references, please type NA.: NA

3. Does the introduction state the objectives of the manuscript in terms that encourage the reader to read on? If not, please explain your answer in the Public Comments section.: Yes

4. How would you rate the organization of the manuscript? Is it focused? Please elaborate with suggestions for reorganization in the Public Comments section.: Satisfactory

6. How is the length of the manuscript? If changes are suggested, please make explicit recommendations in the Public Comments section.: About right

5. Please rate the readability of the manuscript. Explain your rating under Public Comments below.: Readable - but requires some effort to understand

7. Should the supplemental material be included? (Click on the Supplementary Files icon to view files): Does not apply, no supplementary files included

8. If yes to 7, should it be accepted:

Please rate the manuscript overall. Explain your choice.: Good